

Justify your answers and show your work.

(a) Find the inverse on the specified interval express it in the form $y = f^{-1}(x)$. (b) State the range and domain of $f^{-1}(x)$. (c) Graph $f(x)$ and $f^{-1}(x)$. [2 pts each].

1. $f(x) = x^2 - 6x + 9$ for $x \geq 3$

Find the derivative of the inverse of the following function at the specified point on the graph of the inverse function without finding f^{-1} .

2. $f(x) = 3x^5 - 2x^4 + 1$, at $(1, 0)$ (point on f^{-1}).

Use the information about $f(x)$ to answer the following questions. $f(3) = 3$, $f(-4) = \sqrt{5}$, and $f(x)$ has the domain $(-\infty, \infty)$ and range $[\sqrt{5}, \infty)$ [2pts each].

3. Find $f^{-1}(3)$

4. Find $f^{-1}(\sqrt{5})$

5. What is the domain of $f^{-1}(x)$

Perform the indicated operation.

6. $\frac{d}{dx} x^2 e^{-x}$

7. $\frac{d}{dx} \frac{e^x + 3x}{2}$

8. $\frac{d}{dx} (2 \ln x + \sqrt{2})$

9. $\frac{d}{dx} \left(\ln(3x^2) - \frac{x}{5} \right)$

Perform the indicated operation.

10. $\int \left(3e^{-3x} + \frac{1}{x} \right) dx$

11. $\frac{d}{dx} 5^x dx$

12. $\frac{d}{dx} x^{\sqrt{8}}$

13. $\frac{d}{dx} x^{\ln x}$ (use logarithmic differentiation)

Perform the indicated operation.

14. $\int_1^2 2^x$

15. $\int 3xe^{2x} dx$

16. $\int x \ln 2x dx$

17. $\int x^2 \cos 4x dx$